

Problem 15.3.3 (Shankar).

(1) $\hat{J}_x, \hat{J}_y, \hat{J}_z$ are the three components of a vector operator.

Hence we can define

$$J_z = J_1^0; \quad J_x = \frac{J_1^{-1} - J_1^{+1}}{\sqrt{2}}; \quad J_y = \frac{i}{\sqrt{2}} (J_1^{-1} + J_1^{+1})$$

Now consider $\langle \alpha j m | J_1^q | \alpha' j' m' \rangle$. Using Wigner-Eckart thm.

$$\langle \alpha j m | J_1^q | \alpha' j' m' \rangle = \langle \alpha j || J_1 || \alpha' j' \rangle \langle j m | j' m', 1 q \rangle \quad \text{--- (1)}$$

But we can evaluate the L.H.S for $q=0$! We see

$$\langle \alpha j m | J_1^0 | \alpha' j' m' \rangle = \delta_{\alpha\alpha'} \delta_{jj'} \delta_{mm'} m \hbar$$

Further if we set $m=m'=j$ we get.

$$\langle \alpha j j | J_1^0 | \alpha' j' j \rangle = \delta_{\alpha\alpha'} \delta_{jj'} j \hbar \quad \text{--- (2)}$$

But from Wigner Eckart theorem eq. (1) we see

$$\langle \alpha j j | J_1^0 | \alpha' j' j \rangle = \langle \alpha j || J_1 || \alpha' j' \rangle \langle j j | j' j, 1 0 \rangle \quad \text{--- (3)}$$

Comparing eq. (2) and (3) we can deduce $j'=j$ for L.H.S to be non-zero. Setting

thus we get-

$$\langle \alpha j || J_1 || \alpha' j' \rangle \underbrace{\langle j j | j j, 1 0 \rangle}_{\text{we are given } \sqrt{j(j+1)}} = \delta_{jj'} \delta_{\alpha\alpha'} j \hbar$$

$$\therefore \langle \alpha j || J_1 || \alpha' j' \rangle = \delta_{jj'} \delta_{\alpha\alpha'} \sqrt{j(j+1)} \hbar$$

(2) We are given that $\vec{J} \cdot \vec{A} = J_z A_z + \frac{1}{2} (J_- A_+ + J_+ A_-)$

where $A_+ = A_x + iA_y = -\sqrt{2}A_1'$
 $A_- = A_x - iA_y = +\sqrt{2}A_1'^{-1}$ and $A_z = A_1^0$

$\therefore$ We see

$$\vec{J} \cdot \vec{A} = J_z A_1^0 - \frac{1}{\sqrt{2}} (J_- A_1' - J_+ A_1'^{-1})$$

Let us now calculate $\langle \alpha' j m' | \vec{J} \cdot \vec{A} | \alpha j m \rangle$. This is

$$\langle \alpha' j m' | \left\{ J_z A_1^0 - \frac{1}{\sqrt{2}} (J_- A_1' - J_+ A_1'^{-1}) \right\} | \alpha j m \rangle$$

$$= m' \hbar \langle \alpha' j m' | A_1^0 | \alpha j m \rangle - \frac{\sqrt{(j-m')(j+m'+1)}}{\sqrt{2}} \hbar \langle \alpha' j m'+1 | A_1' | \alpha j m \rangle$$

$$+ \frac{\sqrt{(j+m')(j-m'+1)}}{\sqrt{2}} \hbar \langle \alpha' j m'-1 | A_1'^{-1} | \alpha j m \rangle$$

$$= \left\{ m' \hbar \langle j m' | j m, 10 \rangle - \frac{\sqrt{(j-m')(j+m'+1)}}{\sqrt{2}} \hbar \langle j m'+1 | j m, 11 \rangle \right.$$

$$\left. + \frac{\sqrt{(j+m')(j-m'+1)}}{\sqrt{2}} \hbar \langle j m'-1 | j m, 1-1 \rangle \right\}$$

$$\langle \alpha' j || A_1 || \alpha j \rangle \quad \text{--- (4)}$$

$c = \{ \dots \}$ ← quantity inside above

Now $\langle j m' | J_1^q | j m \rangle = \underbrace{\langle j || J_1 || j \rangle}_{\hbar \sqrt{j(j+1)}} \langle j m' | j m, 1q \rangle$ ← from part (1) of our problem

$$\therefore \langle j m' | j m, 1q \rangle = \frac{\langle j m' | J_1^q | j m \rangle}{\hbar \sqrt{j(j+1)}} \quad \text{--- (5)}$$

Substituting in eq. (4) we get

$$c = \left\{ m' \frac{\langle j m' | J_1^0 | j m \rangle}{\hbar \sqrt{j(j+1)}} - \frac{\sqrt{(j-m')(j+m'+1)}}{\sqrt{2}} \frac{\langle j(m'+1) | J_1^1 | j m \rangle}{\hbar \sqrt{j(j+1)}} \right. \\ \left. + \frac{\sqrt{(j+m')(j-m'+1)}}{\sqrt{2}} \frac{\langle j(m'-1) | J_1^1 | j m \rangle}{\hbar \sqrt{j(j+1)}} \right\}$$

Now $J_1^1 = -\frac{J^-}{\sqrt{2}}$ $J_1^{-1} = \frac{J^+}{\sqrt{2}}$. substituting this and using raising + lowering operators we get-

$$c = \frac{\hbar}{\sqrt{j(j+1)}} \left\{ m'^2 + \frac{1}{2} (j-m')(j+m'+1) + \frac{1}{2} (j+m')(j-m'+1) \right\} \delta_{mm'}$$

$$c = \hbar \sqrt{j(j+1)} \delta_{mm'}$$

(3) From Wigner-Eckart theorem we see

$$\langle \alpha' j m' | A_1^q | \alpha j m \rangle = \langle \alpha' j || A_1 || \alpha j \rangle \frac{\langle j m' | j m, 1 q \rangle}{\sqrt{j(j+1)}}$$

Using the result of part (2) we see that the R.H.S is

$$= \frac{\langle \alpha' j m | \vec{J} \cdot \vec{A} | \alpha j m \rangle \langle j m' | j m, 1 q \rangle}{\hbar \sqrt{j(j+1)}}$$

But we also found $\frac{\langle j m' | J_1^q | j m \rangle}{\hbar \sqrt{j(j+1)}} = \langle j m' | j m, 1 q \rangle$

from eq. (5). substituting we see the R.H.S is

$$\frac{\langle \alpha' j m | \vec{J} \cdot \vec{A} | \alpha j m \rangle \langle j m' | J_1^q | j m \rangle}{\hbar^2 j(j+1)}$$